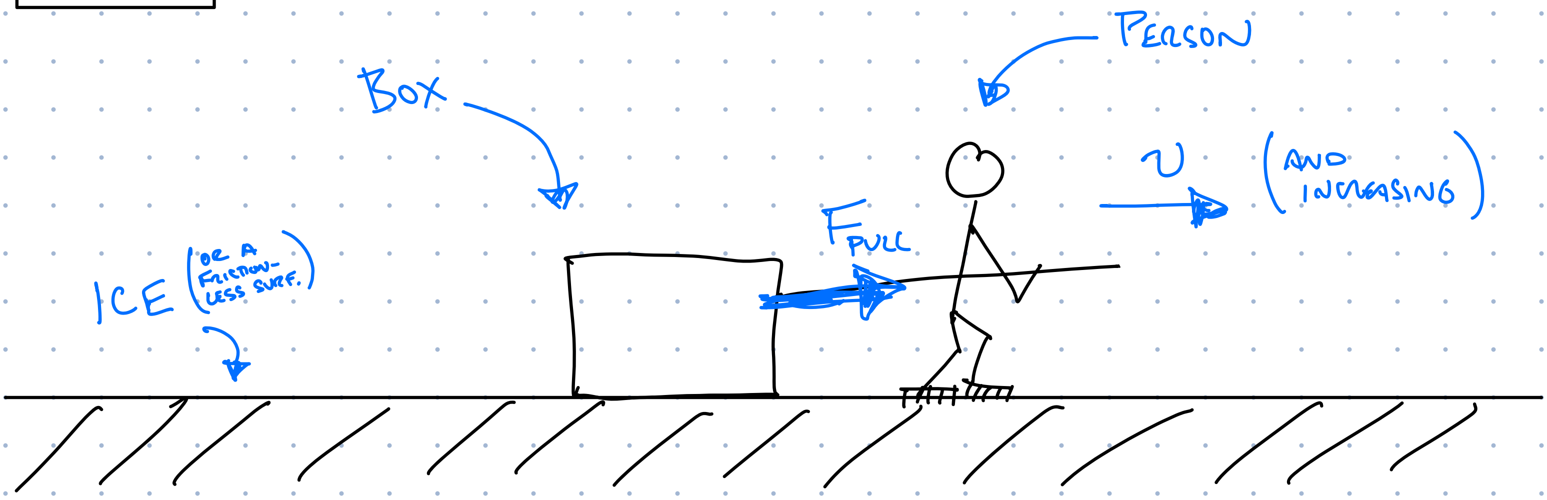
Q 2(a)



<u>Situation</u>: Person pulling box across ice
(Pull force does work on box → increase KE)

<u>Time period of consideration</u>: Just after starting to pull

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Objects, Energies, and Energy Flows

Person	<u>External Energy</u> KE ↑ ④
<u>Internal Energy</u> Chemical Energy ↓ (use of ATP from food) ⑤	<u>Total Energy</u> ↓ ③

② Work →

⑤ If the person's total ↓ but external ↑ then their "internal" must be going <u>down</u>. This is the use of ATP to move muscles.

Box	<u>External Energy</u> KE ↑ ①
<u>Internal Energy</u>	<u>Total Energy</u> ↑ ①

⑥ The floor/ice's energy is <u>unchanged</u> because (without friction) there is no thermal energy generate (and no "motion")

Floor/ ICE	<u>External Energy</u>
<u>Internal Energy</u>	<u>Total Energy</u> ⑥ ↔ (unchanged)

③ If energy is leaving the person, by the work, then "total" is <u>going down</u>.

④ But as person pulls the box, they <u>speed up</u>, so their "external" <u>increases</u>.

② The energy increase of the box must come from the person. So energy is transferred person → box by "work"

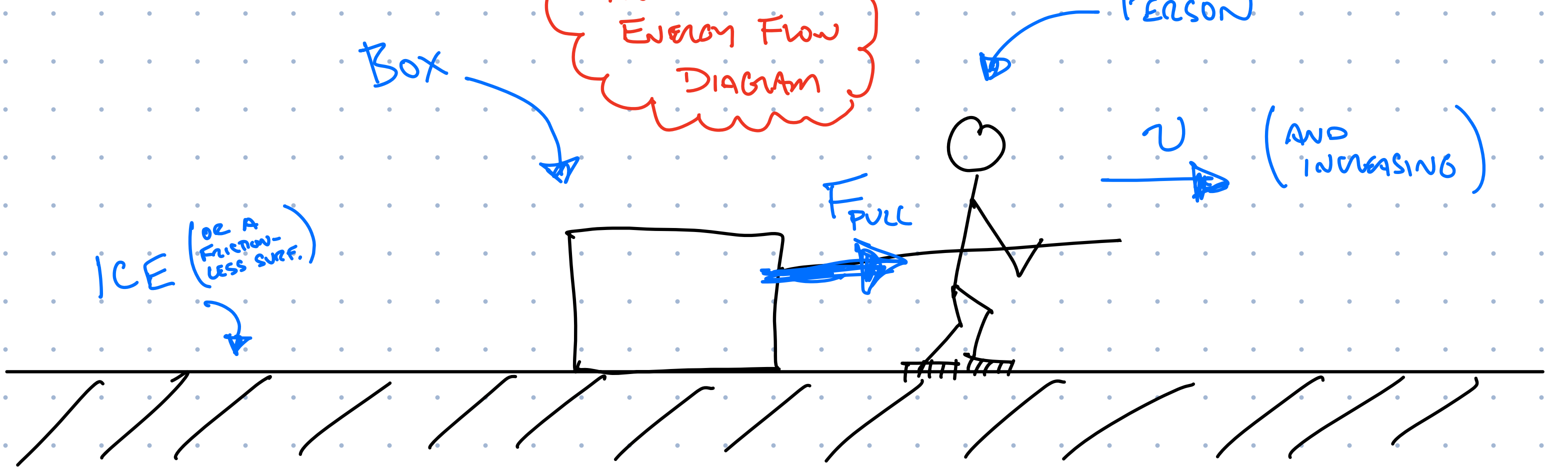
① The work done by person:

$$W = F_{pull} \times Dist$$

is increasing the kinetic energy of the box (speeding up). Otherwise no change. So "total" is increasing.

Q 2 (a)

NOW WITH "NUMBERS" ON ENERGY FLOW DIAGRAM



SITUATION: PERSON PULLING BOX ACROSS ICE
(PULL FORCE DOES WORK ON BOX → INCREASE KE)

TIME PERIOD OF CONSIDERATION: JUST AFTER STARTING TO PULL

OBJECTS, ENERGIES, AND ENERGY FLOWS

PERSON

EXTERNAL ENERGY: KE ↑ 5000 ④ (PERSON MORE MASSIVE THAN BOX)

INTERNAL ENERGY: CHEMICAL ENERGY ↓ (USE OF ATP FROM FOOD) ⑤ 5100

TOTAL ENERGY: ↓ 100 ③

② WORK 100 (PERSON → BOX)

⑤ IF THE PERSON'S TOTAL ↓ BUT EXTERNAL ↑ THEN THEIR "INTERNAL" MUST BE GOING DOWN. THIS IS THE USE OF ATP TO MOVE MUSCLES.

BOX

EXTERNAL ENERGY: KE ↑ ① 100

INTERNAL ENERGY:

TOTAL ENERGY: ↑ 100 ①

⑥ THE FLOOR/ICE'S ENERGY IS UNCHANGED BECAUSE (WITHOUT FRICTION) THERE IS NO THERMAL ENERGY GENERATE (AND NO "MOTION")

FLOOR/ICE

EXTERNAL ENERGY:

INTERNAL ENERGY:

TOTAL ENERGY: ⑥ ↔ (UNCHANGED)

③ IF ENERGY IS LEAVING THE PERSON, BY THE WORK, THEN "TOTAL" IS GOING DOWN.

④ BUT AS PERSON PULLS THE BOX, THEY SPEED UP, SO THEIR "EXTERNAL" INCREASES.

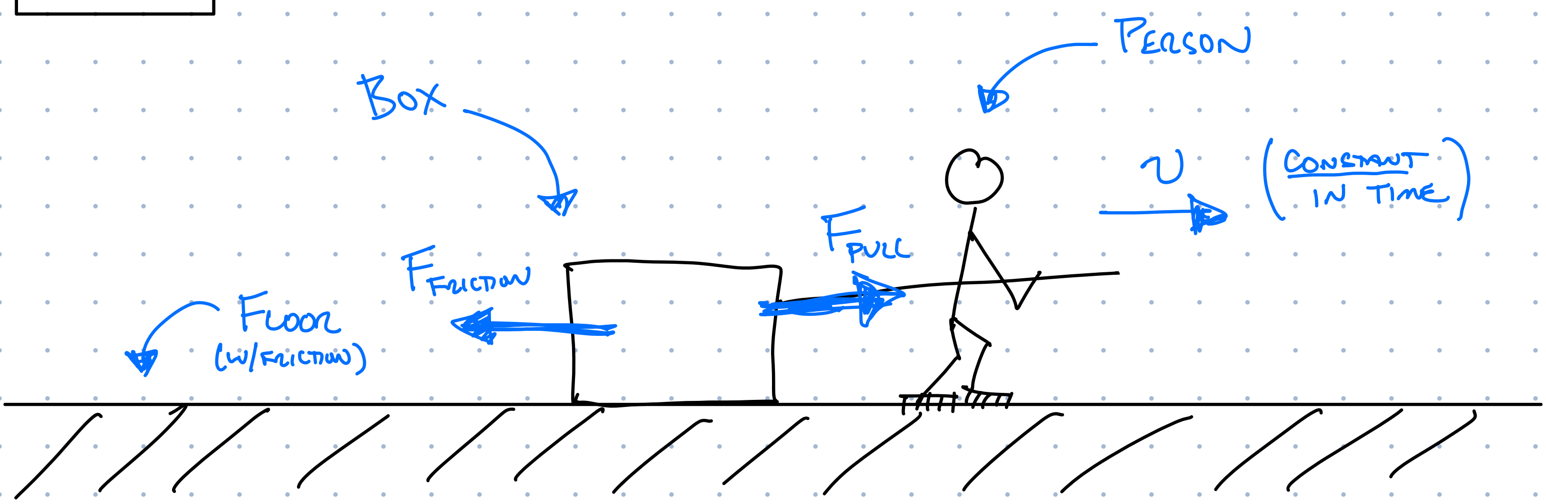
② THE ENERGY INCREASE OF THE BOX MUST COME FROM THE PERSON. SO ENERGY IS TRANSFERRED PERSON → BOX BY "WORK"

① THE WORK DONE BY PERSON:

$$W = F_{PULL} \times DIST$$

IS INCREASING THE KINETIC ENERGY OF THE BOX (SPEEDING UP). OTHERWISE NO CHANGE. SO "TOTAL" IS INCREASING.

Q 2 (b)



Situation: Person Pulling Box Across Floor, With Friction

(Work done by Pulling and Friction Forces Cancel $\Rightarrow \Delta K = 0$
$\Rightarrow v = \text{const}$)

Time Period of Consideration: Any time you like... It's in equilibrium and unchanging over time!

Objects, Energies, and Energy Flows

Work By Pull ③ (Person → Box)

Person	External Energy: KE ↔ ⑤
Internal Energy: Chemical ↓ ⑥	Total Energy: ↓ ④ (Only Energy Output)

Box	External Energy: KE ↔ ① (Const Speed)
Internal Energy: Thermal ↑ (Heats up due to Fric)	Total Energy: ↑

Frictional Work (Box ↔ Floor)

Floor	External Energy: ↔
Internal Energy: Thermal ↑	Total Energy: ↑

⑩ Frictional work transfers energy from Box to Floor and from Floor to Box

⑦ As Box drags on Floor, the Floor heats up, so Internal Increases.

⑧ Floor is not in motion, so KE, and all External, is Unchanged

⑨ So Total for Floor is Increased.

③ Work done by Pulling Force is putting energy into Box, from the Person.

④ So Total for Person is Decreasing,

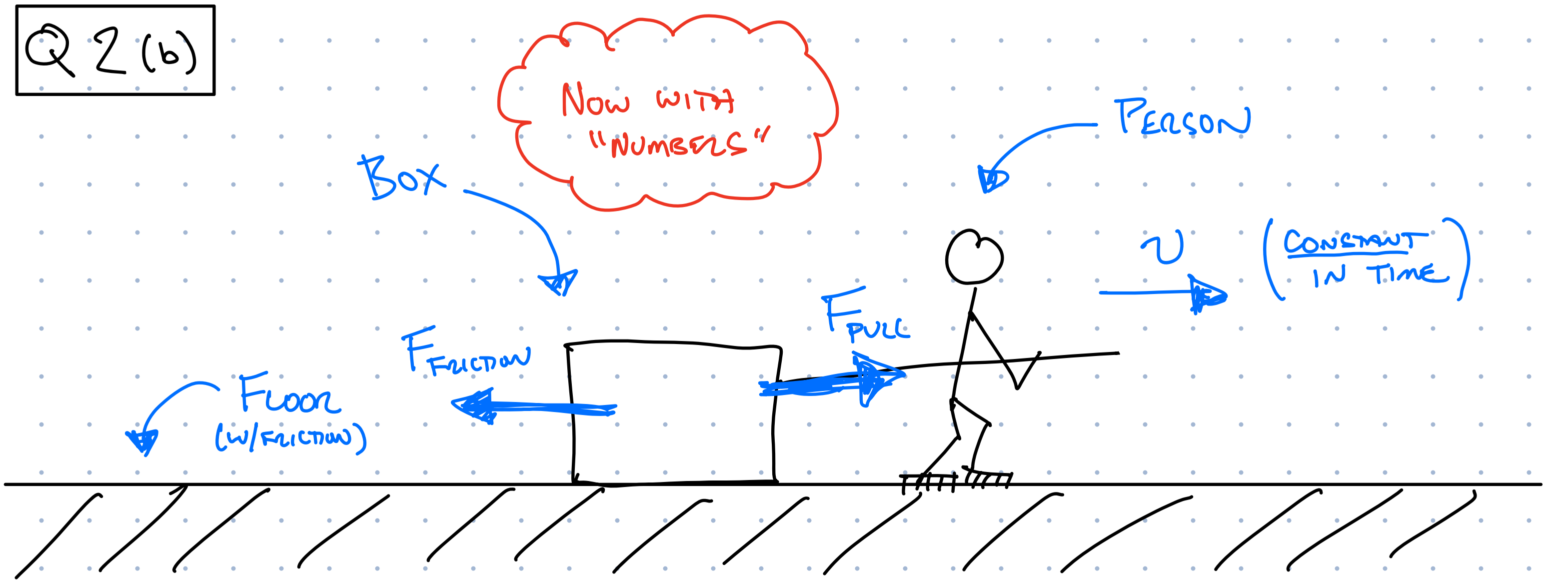
⑤ Since $v = \text{const}$, KE is Unchanged.

⑥ Internal Energy (Chemical Energy of ATP) is Decreasing.

① The Box is moving at a const. speed so KE is Unchanged.

② But it drags on Floor with Friction so Internal (Thermal) Energy is Increasing $\Rightarrow$ Total Increasing

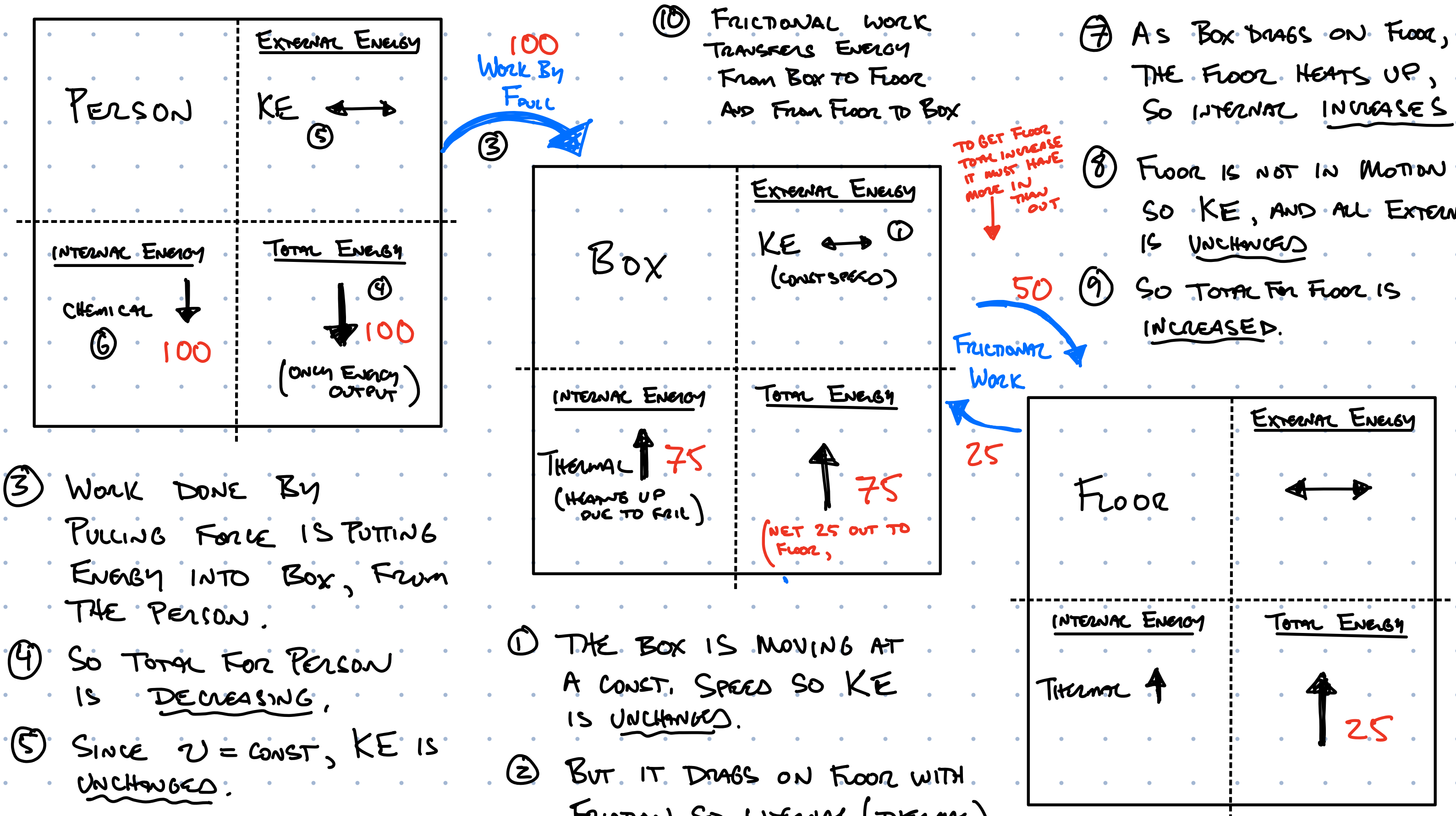
Q 2 (b)



SITUATION: Person Pulling Box Across Floor, With Friction
(Work done by pulling and friction forces cancel $\Rightarrow \Delta K = 0$
$\Rightarrow v = \text{const}$)

TIME PERIOD OF CONSIDERATION: Any time you like... It's in equilibrium and unchanging over time!

Objects, Energies, and Energy Flows



⑩ Frictional work transfers energy from Box to Floor and from Floor to Box

⑦ As Box drags on Floor, the Floor heats up, so internal increases.

⑧ Floor is not in motion, so KE, and all external is unchanged

⑨ So total for Floor is increased.

③ Work done by pulling force is putting energy into Box, from the person.

④ So total for person is decreasing.

⑤ Since $v = \text{const}$, KE is unchanged.

⑥ Internal energy (chemical energy of ATP) is decreasing.

① The Box is moving at a const. speed so KE is unchanged.

② But it drags on floor with friction so internal (thermal) energy is increasing $\Rightarrow$ total increasing